

9. Array Minimise the maximum difference between heights [V.IMP]

class Solution {

public:

int getMinDiff(vector<int> &arr, int k) {

int n = arr.size();

sort(arr.begin(), arr.end());

int diff = arr[n-1] - arr[0]; // initial difference

int smallest = arr[0] + k;

int largest = arr[n-1] - k;

for(int i = 1; i < n; i++) {

int min_height = min(smallest, arr[i] - k);

int max_height = max(largest, arr[i-1] + k);

if(min_height < 0) continue; // skip if negative

diff = min(diff, max_height - min_height);

}

return diff;

}

};

The screenshot shows a C++ IDE with the following components:

- Output Window:** Displays "Problem Solved Successfully" with a green checkmark. It also shows "Test Cases Passed: 1115 / 1115", "Attempts: Correct / Total: 1 / 1", "Accuracy: 100%", "Points Scored: 4 / 4", and "Time Taken: 0.22".
- Code Editor:** Contains the C++ code for the solution, which is a class `Solution` with a public method `getMinDiff`. The code sorts the array and calculates the minimum difference between heights after adjusting by `k`.
- Problem List:** At the bottom, there are three buttons: "Minimum Jumps", "A difference of values and indexes", and "Minimize the Heights I".